College of Engineering – Department of Civil Engineering

CURRICULUM

Bachelor of Science in Transportation Systems Engineering

Academic Year 2025-2026

Reference CMOs: CMO No. 86, S. 2017, CMO No. 20 s. 2013, CMO No. 39 s. 2021, and CMO No. 40 s. 2021

Curriculum Description

The increasing transportation demand for systematized mobility of people, goods and services as part of rural development and urbanization all over the country indicates the need to increase specialists for transportation sector. The offering of Bachelor of Science in Transportation Systems Engineering program in the university will provide skilled professionals that will take the lead in solving various transportation challenges. Bachelor of Science in Transportation Systems Engineering program provides strong foundation in transportation planning/management, traffic engineering, and intelligent transport systems. The program will adopt outcome-based education (OBE) framework with flipped classroom and other blended learning pedagogies.

Program Education Objectives

The transportation systems engineering alumni three to five years after graduation shall:
PEO 1: Specialist. Practiced as specialist in solving complex transportation systems engineering problems leading to improvements and innovations, while taking into consideration the environmental, social, and economical requirements.
PEO 2: Professionalism and Leadership. Assumed leadership position in industry, academe, government, or private sector within consideration to social and ethical responsibility.
PEO 3: Lifelong Learning. Engaged in lifelong learning through further studies, research, certifications, promotions, and other personal and professional development activities.

Student Outcomes

The following skills, knowledge, and values are expected to be attained by students as they progress through the program:
SO1 Discipline Knowledge. Ability to apply mathematics, sciences and principles of engineering to solve complex transportation systems engineering problems;
SO2 Investigation. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions;
SO3 Design/Development of Solutions. Design solution, system, components, processes, exhibiting improvements/innovations, that meet specified needs with appropriate consideration for public health and safety, cultural, societal, economical, ethical, environmental and sustainability issues.
SO4 Leadership and Teamwork. Function effectively as a member of a leader on a diverse team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
SO5 Problem Analysis. Identify, formulate, and solve complex transportation systems engineering problems by applying principles of engineering, science, and mathematics;
SO6 Ethics and Professionalism. Apply ethical principles and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, environmental, and societal contexts.
SO7 Communication. Communicate effectively on complex transportation systems engineering activities with the community, and the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions;

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SO8 Environment and Sustainability. Recognize the impact of professional engineering solutions in societal, global, and environmental contexts and demonstrate knowledge of and need for sustainable development;

SO9 Lifelong Learning. Recognize the need for, and ability to engage in independent and life-long learning in the broadest context of technological change.

SO10 The Engineer and Society. Apply reasoning based on contextual knowledge to assess societal, health, safety, legal, cultural, contemporary issues, and the consequent responsibilities relevant to professional engineering practices.

SO11 Modern Tool Usage. Apply appropriate techniques, skills, and modern engineering and IT tools to complex transportation systems engineering activities;

SO12 Project Management and Finance. Demonstrate knowledge and understanding of engineering management and financial principles as member or a leader of a team to manage projects in multidisciplinary settings and identify opportunities of entrepreneurship.

SO13 Social and National Responsibility. Apply acquired engineering knowledge and skills in addressing community problems that contributes to national development.

CURRICULUM COMPONENTS

Classification / Field / Course	Number of Hours per Week		Credit Units
	Lec	Lab	
I. TECHNICAL COURSES			
A. Mathematics			
Differential Calculus	5	0	5
Integral Calculus	5	0	5
Differential Equations	3	0	3
Engineering Data Analysis	3	0	3
Numerical Solutions to Transportation Engineering	3	0	3
Subtotal	**19**	**0**	**19**
B. Natural/ Physical Sciences			
Chemistry for Engineers	3	3	4
Physics for Engineers 1	3	3	4
Physics for Engineers 2	3	3	4
Subtotal	**9**	**9**	**12**
C. Basic Engineering Sciences			
Introduction to Engineering	0	3	1
Engineering Drawing	0	3	1
Computer Programming	0	3	1
Computer Aided Design	0	3	1
Engineering Economics	3	0	3
Technopreneurship	3	0	3
Engineering Management	2	0	2
Statics of Rigid Bodies	3	0	3
Dynamics of Rigid Bodies	2	0	2
Transportation Engineering Orientation	2	0	2
Environmental Science and Engineering	3	0	3
Basic Occupational Safety and Health	3	0	3
Research Methods	3	0	3
Subtotal	**24**	**12**	**28**
D. Allied Courses			
Fundamentals of Electronics Communications	3	0	3

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Navigational Aids	2	0	2
Strength of Materials	4	0	4
Hydrology	3	0	3
Construction Materials and Testing	2	3	3
Hydraulics	4	0	4
Geotechnical Engineering I: Soil Mechanics	3	3	4
General Surveying I	2	3	3
General Surveying II	2	3	3
Subtotal	**25**	**12**	**29**
E. Professional Courses			
GIS Application for Transportation Engineering	0	3	1
Geometric Design and Highway Safety	4	0	4
Fundamentals of Traffic Engineering	3	0	3
Fundamentals of Airport Engineering and Management	3	0	3
Introduction to Marine Transportation	3	0	3
Fundamentals of Railway Engineering	3	0	3
Traffic Engineering Laboratory	3	3	4
Transportation Networks and Optimization	2	3	3
Pavement Design and Testing	2	3	3
Railways Signaling and Management	3	0	3
Mass Transit Engineering	3	0	3
Planning and Design of Transportation Infrastructures	3	0	3
Transportation System Analysis and Planning	3	0	3
Traffic Impact Assessment Methods	3	0	3
Transportation Project Management	3	0	3
Transportation Engineering Economics and Finance	3	0	3
Transportation Laws and Policy Development	3	0	3
Transportation Engineering Design Project 1	1	3	2
Transportation Engineering Design Project 2	0	6	2
Subtotal	**48**	**21**	**55**
F. On-the-Job Training			
On-the-Job Training	320 hrs		4
Subtotal	**-**	**-**	**4**
TOTAL TECHNICAL COURSES	**125**	**54**	**147**
II. NON-TECHNICAL COURSES			
A. General Education Courses			
Understanding the Self	3	0	3
Mathematics for the Modern World	3	0	3
The Contemporary World	3	0	3
Readings in Philippine History	3	0	3
Purposive Communication	3	0	3
Ethics	3	0	3
Art Appreciation	3	0	3
Science, Technology, and Society	3	0	3
Subtotal	**24**	**0**	**24**

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B. General Education Electives and Mandated Courses			
Technical Writing	3	0	3
Living in the IT Era	3	0	3
The Entrepreneurial Mind	3	0	3
Life and Works of Rizal	3	0	3
Subtotal	**12**	**0**	**12**
C. Physical Education Toward Health and Fitness			
Movement Competency Training	2	0	2
Exercise-based Fitness Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities	2	0	2
Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities	2	0	2
Subtotal	**8**	**0**	**8**
D. National Service Training Program			
National Service Training Program 1	3	0	3
National Service Training Program 2	3	0	3
Subtotal	**6**	**0**	**6**
TOTAL NON-TECHNICAL COURSES	**50**	**0**	**50**
GRAND TOTAL	**175**	**54**	**197**

SUMMARY

Courses	Number of Units
I. TECHNICAL COURSES	
A. Mathematics	**19**
B. Natural/Physical Sciences	**12**
C. Basic Engineering Sciences	**28**
D. Allied Courses	**29**
E. Professional Courses	**55**
F. On the Job Training	**4**
II. NON-TECHNICAL COURSES	
A. General Education Courses	**24**
B. General Education Electives and Mandate Courses	**12**
C. Physical Education Toward Health and Fitness	**8**
D. National Service Training Program	**6**
GRAND TOTAL	**197**

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CURRICULUM STRUCTURE

BACHELOR OF SCIENCE IN TRANSPORTATION SYSTEMS ENGINEERING

FIRST YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Lec**	**Lab**	**Credit Units**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
SCI 401	Chemistry for Engineers	3	3	4			
GEd 105	Readings in the Philippine History	3	0	3			
GEd 108	Art Appreciation	3	0	3			
Math 401	Differential Calculus	5	0	5			
GEd 104	The Contemporary World	3	0	3			
CpE 401	Computer Programming	0	3	1			
ENGG 401	Introduction to Engineering	0	3	1			
PATHFit 1	Movement Competency Training	2	0	2			
NSTP 111	National Service Training Program 1	3	0	3			
	Total	**22**	**9**	**25**			
FIRST YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Lec**	**Lab**	**Credit Units**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
SCI 403	Physics for Engineers 1	3	3	4	Math 401, SCI 401		
GEd 102	Mathematics in the Modern World	3	0	3			
GEd 106	Purposive Communication	3	0	3			
Ged 101	Understanding the Self	3	0	3			
Math 402	Integral Calculus	5	0	5	Math 401	SCI 403	
ENGG 402	Engineering Drawing	0	3	1	ENGG 401		
Ged 109	Science, Technology and Society	3	0	3			
PATHFit 2	Exercise-based Fitness Activities	2	0	2	PATHFit 1		
NSTP 112	National Service Training Program 2	3	0	3	NSTP 111		
	Total	**25**	**6**	**27**			
FIRST YEAR							
MIDTERM							

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Course Code	Course Title	Lec	Lab	Credit Units	Prerequisite(s)	Co-requisite(s)	Category
SCI 404	Physics for Engineers 2	3	3	4	SCI 403, MATH 402		
GEd 103	Life and Works of Rizal	3	0	3			
GEd 107	Ethics	3	0	3			
	Total	**9**	**3**	**10**			
SECOND YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Lec**	**Lab**	**Credit Units**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
GedE 103	The Entrepreneurial Mind	3	0	3			
Math 403	Engineering Data Analysis	3	0	3	Math 402		
Math 404	Differential Equations	3	0	3	Math 402		
ENGG 408	Dynamics of Rigid Bodies	2	0	2	SCI 403, Math 402		
ENGG 409	Statics of Rigid Bodies	3	0	3	SCI 403, Math 402		
GE 402	General Surveying I	2	3	3	ENGG 402		
PATHFit 3	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and Adventure Activities	2	0	2	PATHFit 2		
ENGG 403	Computer-Aided Design	0	3	1			
TE 401	Transportation Engineering Orientation	2	0	2			
Total		**20**	**6**	**22**			
SECOND YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Lec**	**Lab**	**Credit Units**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
GedE 101	Technical Writing	3	0	3			
CE 402	Strength of Materials	4	0	4	ENGG 407		
CE 410	Hydraulics	4	0	4	ENGG 407	CE 402	
CE 405	Hydrology	3	0	3		CE 410	
CE 406	Construction Materials and Testing	2	3	3		CE 402	
GE 404	General Surveying II	2	3	3	GE 402		
ECE 430	Fundamentals of Electronics Communication	3	0	3	SCI 404		
TE 402	Numerical Solutions to Transportation Engineering	3	0	3	MATH 404		
PATHFit 4	Choice of Dance, Sports, Martial Arts, Group Exercise, Outdoor and	2	0	2	PATHFit 3		

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	Adventure Activities						
	Total	**26**	**6**	**28**			
THIRD YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Lec**	**Lab**	**Credit Units**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
TE 403	GIS Application for Transportation Engineering	0	3	1	GE 404		
TE 404	Geometric Design and Highway Safety	4	0	4	GE 404		
TE 405	Fundamentals of Traffic Engineering	3	0	3	ENGG 408		
TE 406	Fundamentals of Airport Engineering and Management	3	0	3	ENGG 408	ECE 431	
TE 407	Introduction to Marine Transportation	3	0	3	CE 410		
TE 408	Fundamentals of Railway Engineering	3	0	3	GE 404		
CE 411	Geotechnical Engineering I: Soil Mechanics	3	3	4	CE 402		
ECE 431	Navigational Aids	2	0	2	ECE 430		
	Total	**20**	**6**	**23**			
THIRD YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Lec**	**Lab**	**Credit Units**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
TE 409	Traffic Engineering Laboratory	3	3	4	TE 405		
TE 410	Transportation Networks and Optimization	2	3	3	TE 405	TE 409	
TE 411	Pavement Design and Testing	2	3	3	CE 411, TE 404		
TE 412	Railways Signaling and Management	3	0	3	ECE 431, TE 408		
TE 413	Mass Transit Engineering	3	0	3			
TE 414	Planning and Design of Transportation Infrastructures	3	0	3	TE 404		
TE 415	Transportation System Analysis and Planning	3	0	3	TE 404	TE 414	
TE 416	Traffic Impact Assessment Methods	3	0	3	TE 405	TE 409	
Total		**22**	**9**	**25**			
THIRD YEAR							
MIDTERM							

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Course Code	Course Title	Lec	Lab	Credit Units	Prerequisite(s)	Co-requisite(s)	Category
ENGG 416	Research Methods	3	0	3	GEdE 101		
ENGG 404	Engineering Economics	3	0	3	MATH 402		
GEdE 102	Living in the IT Era	3	0	3			
Total		**9**	**0**	**9**			
FOURTH YEAR							
FIRST SEMESTER							
Course Code	**Course Title**	**Lab**	**Lec**	**Credit Units**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
TE 417	Transportation Project Management	3	0	3	TE 416, TE 414		
TE 418	Transportation Engineering Economics and Finance	3	0	3	ENGG 404		
TE 419	Transportation Laws and Policy Development	3	0	3			
TE 420	Transportation Engineering Design Project 1	1	3	2	ENGG 416, TE 414, TE 415, TE 416		
ENGG 406	Engineering Management	2	0	2			
ENGG 405	Technopreneurship	3	0	3			
ENGG 411	Basic Occupational Safety and Health	3	0	3			
	Total	**18**	**3**	**19**			
FOURTH YEAR							
SECOND SEMESTER							
Course Code	**Course Title**	**Lec**	**Lab**	**Credit Units**	**Prerequisite(s)**	**Co-requisite(s)**	**Category**
ENGG 413	Environmental Science and Engineering	3	0	3			
TE 421	Transportation Engineering Design Project 2	0	6	2	TE 421		
ENGG 417	On-the-Job Training	320		4	Graduating standing		
	Total	**3**	**6**	**9**			
GRAND TOTAL		**175**	**54**	**197**			